ROW-REDUCED ECHELON FORM
&
VECTOR SPACES

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INTRODUCTION

- A lot of matrix mathematics are handled better when using the row-reduced echelon form. Some of these, include finding the inverses of matrices, solving a series of equations, finding column and null spaces and many more.
- The row-reduced echelon form was derived from method of finding the inverse matrix using the gaussian-elimination method. Hence, the row-reduced echelon form is also known as the Gauss-Elimination method.
- In this session, we will not bother to look at finding inverses or solving equations using the row-reduced echelon form in detail, we will rather look at the methodology, and then proceed to column and null spaces.

ROW-REDUCED ECHELON FORM

- There are two things occurring here, the echelon form and the reduced echelon form.
- In Row echelon form, the matrix has to satisfy the following properties.
    1. All non-zero rows are above any rows of all zeros.
    2. Each leading entry (i.e. left most nonzero entry) of a row is in a column to the right of the leading entry of the row above it.
    3. All entries in a column below a leading entry are zero.
- In general, we will define some of the general forms of matrices in row echelon form.

ROW-REDUCED ECHELON FORM

- Below are the general forms of the row echelon form.

a) $\begin{bmatrix} \blacksquare & * & * & * & * \\ 0 & \blacksquare & * & * & * \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

c.) $\begin{bmatrix} 0 & \blacksquare & * & * & * & * & * \\ 0 & 0 & 0 & \blacksquare & * & * & * \\ 0 & 0 & 0 & 0 & \blacksquare & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & \blacksquare \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\blacksquare$ Non-Zero entry

$*$ Zero or non-zero entry

b) $\begin{bmatrix} \blacksquare & * & * \\ 0 & \blacksquare & * \\ 0 & 0 & \blacksquare \\ 0 & 0 & 0 \end{bmatrix}$

NOTE: Make sure that we are only having zeros below every leading non-zero entry in a column. And to the left of a leading non-zero entry in a given row, we also have zeros.

ROW-REDUCED ECHELON FORM

- In row-reduced echelon form, we make more changes to the leading entries and their respective columns.
- So, to the first 3 properties of row echelon form, add the following properties.
  4. The leading entry in each nonzero row is 1.
  5. Each leading 1 is the only nonzero entry in its column.
- In general, $\begin{bmatrix} 1 & * & 0 & 0 & * & * & 0 & 0 & * \\ 0 & 0 & 1 & 0 & * & * & 0 & 0 & * \\ 0 & 0 & 0 & 1 & * & * & 0 & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \end{bmatrix}$ is in row-reduced echelon form.

ROW-REDUCED ECHELON FORM

- In this session we will deal more with the row-reduced echelon form and we will often meet a number of terms. For example, pivot position, pivot and pivot column are the most common terms to be familiar with.
  - **pivot position:** a position of a leading entry in an echelon form of the matrix.
  - **pivot:** a nonzero number that either is used in a pivot position to create 0’s or is changed into a leading 1, which in turn is used to create 0’s in row-reduced echelon form.
  - **pivot column:** a column that contains a pivot position.

ROW-REDUCED ECHELON FORM

- Let's try to identify if matrices are in row-reduced echelon form or not.

1. $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -2 \end{bmatrix}$ This matrix is not in row-reduced echelon form because a row with all 0's is above row(s) with some non-zero entries.

2. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ This matrix in row-reduced echelon form because each column is a pivot column and every leading pivot (leading entry) is a 1.

3. $\begin{bmatrix} 1 & 2 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 & 9 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ *(this matrix is left for your practice)*

FINDING REDUCED ECHELON FORMS OF MATRICES

- In the conversion of a matrix into row-reduced echelon form (RREF), we create an augmented matrix depending on the nature of a calculation we need to make.
- For example, if we want to find the inverse matrix, we augment the given square matrix (A_n) and its identity matrix (I_n) in the form $\{\mathrm{A}_n|\mathrm{I}_n\}$. By converting A_n into RREF, the matrix I_n is converted into the inverse of A_n. In the same way, a system of equations $\mathrm{A}\bar{\mathrm{x}} = \bar{\mathrm{b}}$, we augment the coefficient matrix A, and the solution vector $\bar{\mathrm{b}}$, in the form $\{\mathrm{A}|\bar{\mathrm{b}}\}$. In RREF we find the column vector $\bar{\mathrm{x}}$.

ROW-REDUCED ECHELON FORM

- To convert a matrix into RREF, we use the following three rules simultaneously.
  1. Interchanging the positions of any two rows.
  2. Multiplying by a non-zero scalar
  3. Adding or subtracting other rows or adding or subtracting the scalar multiples of other rows.
- We should always note that we can’t multiply by 0 and we also can’t add or subtract a scalar. We should also note that any rule we choose is applied on the entire row and not a specific point or pivot.

ROW-REDUCED ECHELON FORM

Example 1: Find the inverse matrix A given that

$$A = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix}$$

- This a 2×2 matrix, thus the augmented matrix will look as follows

$$\left(\begin{array}{cc|cc} 3 & 1 & 1 & 0 \\ 4 & 2 & 0 & 1 \end{array}\right) = (A|I)$$

- The idea here is to convert A into an identity matrix I, and I into A^{-1}

$$\left(\begin{array}{cc|cc} 3 & 1 & 1 & 0 \\ 4 & 2 & 0 & 1 \end{array}\right) \leftarrow \frac{1}{3}R_1$$

We want to make the first point be 1 (pivot). Since no other row in this column has 1, we can't interchange. Hence, a scalar product to row 1

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- After multiplying $\frac{1}{3}$ to the entire row, we get

$$\left(\begin{array}{cc|cc} 1 & \frac{1}{3} & \frac{1}{3} & 0 \\ 4 & 2 & 0 & 1 \end{array}\right) \leftarrow R_2 - 4R_1$$

We must have all zeros below a pivot, 1. Thus, $4R_1$ will work since we can't subtract by a scalar.

$$\left(\begin{array}{cc|cc} 1 & \frac{1}{3} & \frac{1}{3} & 0 \\ 0 & \frac{2}{3} & -\frac{4}{3} & 1 \end{array}\right) \leftarrow \frac{3}{2}R_2$$

We need to create a pivot for row 2 by converting $\frac{2}{3}$ into 1. Scalar multiplication.

$$\left(\begin{array}{cc|cc} 1 & \frac{1}{3} & \frac{1}{3} & 0 \\ 0 & 1 & -2 & \frac{3}{2} \end{array}\right) \leftarrow R_1 - \frac{1}{3}R_2$$

A pivot column must only contain a 1. We must convert $\frac{1}{3}$ to a 0 but we shouldn't disturb the pivot for row 1.

ROW-REDUCED ECHELON FORM

$$\left(\begin{array}{cc|cc} 1 & 0 & -\frac{1}{3} & -\frac{1}{2} \\ 0 & 1 & -2 & \frac{3}{2} \end{array}\right) = (\mathrm{I}|\mathrm{A}^{-1})$$

At this point we have found row-reduced echelon form of A. For square matrices, the RREF is their respective identity matrices.

Therefore, $A^{-1} = \begin{bmatrix} -\frac{1}{3} & -\frac{1}{2} \\ -2 & \frac{3}{2} \end{bmatrix}$; your job is to test this using $A \cdot A^{-1} = I$.

- As mentioned earlier, more of our attention will be on the process of finding the row-reduced echelon form of any given matrix.

ROW-REDUCED ECHELON FORM

- Some basic points that may help in critical situations when we don't know where to start from may be;
  - Change the first entry in row 1 to 1 by either interchanging the positions with any other row that has 1 in column 1 or multiplying by a scalar. Once this is a pivot (1), we must change everything in the column to zeros.
  - Change the last row to zeros or the last entry to the pivot if possible. If not, start with the rows below the first row, defining pivots where possible. Make sure, you finish with the first row unless otherwise, being in a rush may affect the pivot point in column 1.

ROW-REDUCED ECHELON FORM

- Visually, Given $A_{m \times n}$;

Start with changing $a_{1,1}$ to 1 by any rule. Then change everything in column 1 to 0's. 1

Change the last row into all zeros and stop if and only if you have a pivot at any point in that row. 2

Change the columns that have a pivot to zeros one by one where possible. 3

Change to zeros any entry in any column that contains a pivot. 4

$$\begin{pmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,n} \end{pmatrix}$$

Note that this is simply a proposed idea, it still needs proper analysis of any given matrix.

ROW-REDUCED ECHELON FORM

Example 2: Write the matrix $A = \begin{bmatrix} 0 & 3 & -6 & 6 & 4 & -5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 3 & -9 & 12 & -9 & 6 & 15 \end{bmatrix}$ in RREF.

$$\begin{bmatrix} \boxed{0} & 3 & -6 & 6 & 4 & -5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 3 & -9 & 12 & -9 & 6 & 15 \end{bmatrix} \leftarrow R_1 \leftrightarrow R_3$$

We have 0 on a pivot point. We will interchange with row 3 (row 2 will give fractions)

$$\begin{bmatrix} \boxed{3} & -9 & 12 & -9 & 6 & 15 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{bmatrix} \leftarrow \frac{1}{3}R_1$$

Convert the pivot point to 1. If we had interchanged with row 2, this operation could have brought fractions. Always pay attention.

ROW-REDUCED ECHELON FORM

$$\begin{bmatrix} \boxed{1} & -3 & 4 & -3 & 2 & 5 \\ \boxed{3} & -7 & 8 & -5 & 8 & 9 \\ \boxed{0} & 3 & -6 & 6 & 4 & -5 \end{bmatrix} \leftarrow R_2 - 3R_1$$

Making zeros for all terms below a pivot in a pivot column.

$$\begin{bmatrix} 1 & \boxed{-3} & 4 & -3 & 2 & 5 \\ 0 & \boxed{2} & -4 & 4 & 2 & -6 \\ 0 & \boxed{3} & -6 & 6 & 4 & -5 \end{bmatrix} \leftarrow \frac{1}{2}R_2$$

Creating a pivot in second column. Our consideration is on converting 2 to 1.

$$\begin{bmatrix} 1 & \boxed{-3} & 4 & -3 & 2 & 5 \\ 0 & \boxed{1} & -2 & 2 & 1 & -3 \\ 0 & \boxed{3} & -6 & 6 & 4 & -5 \end{bmatrix} \leftarrow R_3 - 3R_2$$

Making zeros for all terms below a pivot in a pivot column. Start with terms below and lastly finish with entries in row 1.

ROW-REDUCED ECHELON FORM

$$\begin{bmatrix} 1 & -3 & 4 & -3 & 2 & 5 \\ 0 & 1 & -2 & 2 & \boxed{1} & -3 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} \leftarrow R_2 - R_3$$

We have a new pivot in row 3. we need to change everything in this column to 0's above 1.

$$\begin{bmatrix} 1 & -3 & 4 & -3 & \boxed{2} & 5 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} \leftarrow R_1 - 2R_3$$

We have now changed all terms in pivot columns in lower rows. Let's do for row 1

$$\begin{bmatrix} 1 & \boxed{-3} & 4 & -3 & 0 & -8 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} \leftarrow R_1 + 3R_3$$

Make sure that all pivot columns have been treated such that all terms that are not pivots are 0's.

ROW-REDUCED ECHELON FORM

- Hence, we finally get to the final row reduced echelon form of the given matrix. And we have;

Pivots

$$A^* = \begin{bmatrix} 1 & 0 & -2 & 3 & 0 & -24 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

- We may also note that every row operation targets at least one entry changing it to either 1 or 0. However, each operation affects the entire row.

ROW-REDUCED ECHELON FORM

- This algorithm provides a method for using row operations to take a matrix to its echelon form. We begin with the matrix in its original form.
    1. Starting from the left, find the first non-zero column. This is the first pivot column, and the position at the top of this column will be the position of the first pivot entry. Switch rows if necessary to place a non-zero number in the first pivot position.
    2. Use row operations to make the entries below the first pivot entry (in the first pivot column) equal to zero.
    3. Ignoring the row containing the first pivot entry, repeat steps 1 and 2 with the remaining rows. Repeat the process until there are no more non-zero rows left.

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Example 3: Find the reduced echelon form of $A = \begin{bmatrix} 1 & 2 & 1 & 3 & 2 \\ 1 & 3 & 6 & 0 & 2 \\ 3 & 7 & 8 & 6 & 6 \end{bmatrix}$

$$\begin{bmatrix} 1 & 2 & 1 & 3 & 2 \\ 1 & 3 & 6 & 0 & 2 \\ 3 & 7 & 8 & 6 & 6 \end{bmatrix} \xrightarrow{R_2 : (R_2 - R_1)} \begin{bmatrix} 1 & 2 & 1 & 3 & 2 \\ 0 & 1 & 5 & -3 & 0 \\ 3 & 7 & 8 & 6 & 6 \end{bmatrix} \xrightarrow{R_3 : (R_3 - 3R_1)}$$

$$\begin{bmatrix} 1 & 2 & 1 & 3 & 2 \\ 0 & 1 & 5 & -3 & 0 \\ 0 & 1 & 5 & -3 & 0 \end{bmatrix} \xrightarrow{R_3 : (R_3 - R_2)} \begin{bmatrix} 1 & 2 & 1 & 3 & 2 \\ 0 & 1 & 5 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

ROW-REDUCED ECHELON FORM

$$\begin{bmatrix} 1 & 2 & 1 & 3 & 2 \\ 0 & 1 & 5 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1:(R_1-2R_2)} \begin{bmatrix} 1 & 0 & -9 & 9 & 2 \\ 0 & 1 & 5 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- In this question, the first column is a non-zero column, hence, it is our first pivot column. Since the first entry was already a 1, nothing can be done here than making everything in this column equal 0.
- In column 2 we already get another pivot in column 2. The process continues until when all the pivot columns have 1 (pivot) and zeros everywhere. In addition, all entries in a row on the left of a 1 (pivot) are zeros.

ROW-REDUCED ECHELON FORM

A. Determine if the given matrices are in reduced echelon form or not.

1. $\begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -1 \end{bmatrix}$

2. $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

3. $\begin{bmatrix} 1 & 3 & -3 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

4. $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}$

NOTE: Remember to describe the properties in your critics.

Understand the reasons why a given matrix is in row-reduced echelon form or not.

B. Find the echelon and reduced echelon forms of the following matrices

1. $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 5 & 6 & 7 \\ 6 & 7 & 8 & 9 \end{bmatrix}$

2. $\begin{bmatrix} -2 & -3 & -2 \\ 3 & -2 & -2 \\ 3 & -2 & -1 \\ -1 & -1 & -2 \end{bmatrix}$

3. $\begin{bmatrix} 1 & 3 & -3 & -3 \\ 0 & 0 & -2 & 2 \end{bmatrix}$

4. $\begin{bmatrix} 1 & 2 & 0 \\ 1 & 3 & 3 \\ -1 & 0 & -1 \\ -3 & 0 & 0 \end{bmatrix}$

5. $\begin{bmatrix} 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -10 & 6 & 12 & 28 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{bmatrix}$

LINEAR SYSTEM OF EQUATIONS

LINEAR SYSTEM OF EQUATIONS

- A system of linear equations consists of a series of linear equations which is converted and solved using matrix mathematics.
- The general form of these equations is $A\bar{x} = \bar{b}$ where A is the matrix of coefficients in the given equations, $\bar{x}$ is the column vector of all the variables available in the equations and $\bar{b}$ is the column vector of the solution associated with given equations.
- We have two types of equations based on the nature of the solutions in the equations as well as the nature of the values of the variables. These include the non-homogenous equations and homogenous equations.

LINEAR SYSTEM OF EQUATIONS

- As a matter of definition; A linear system of equations $A\bar{x} = \bar{b}$ is called homogeneous if $\bar{b} = 0$, and non-homogeneous if $b \neq 0$.
- Notice that $\bar{x} = 0$ is always solution of the homogeneous equations. That is, $x_1 = 0,\ x_2 = 0, \ldots\ x_n = 0$. This solution is called the trivial solution.
- If there are other solutions, they are called nontrivial solutions. Because a homogeneous linear system always has the trivial solution, there are only two possibilities for its solutions:
  - The system has only the trivial solution.
  - The system has infinitely many solutions in addition to the trivial solution

LINEAR SYSTEM OF EQUATIONS

- In nontrivial cases, we tend to find a non-zero vector that satisfies the equation $A\bar{x} = 0$. The homogeneous equation $A\bar{x} = 0$ has a nontrivial solution if and only if the equation has at least one free variable.
- For non-homogenous equations $A\bar{x} = \bar{b}$, we have various forms and their own solutions. The other methods for solving the equations $A\bar{x} = \bar{b}$ like the inverse method and Cramer's rule are mostly applicable if and only if the coefficient matrix A is a non-singular square matrix ($\det A \neq 0$).
- If the coefficient matrix is not a square matrix there is a slight difference in the nature of the solutions.

LINEAR SYSTEM OF EQUATIONS

- Consider the general for a system of linear equations and their respective augmented matrix when using row-reduced echelon form.

Definition: Augmented matrix of a system of linear equations

Given the following set of linear equations

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$
$$\vdots$$
$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m$$

$$\left(\begin{array}{ccc|c} a_{11} & \cdots & a_{1n} & b_1 \\ \vdots & \ddots & \vdots & \vdots \\ a_{m1} & \cdots & a_{mn} & b_m \end{array}\right)$$

LINEAR SYSTEM OF EQUATIONS

Example 4: Solve the linear equations: $x_1 + 4x_2 + 3x_3 = 11$

$$2x_1 + 10x_2 + 7x_3 = 27$$

$$x_1 + x_2 + 2x_3 = 5$$

- The augmented matrix is

$$\left(\begin{array}{ccc|c} 1 & 4 & 3 & 11 \\ 2 & 10 & 7 & 27 \\ 1 & 1 & 2 & 5 \end{array}\right)$$

- To carry the augmented matrix to reduced echelon form, we will use the method discussed earlier (slide 19). Notice that the first column is non-zero, so this is our first pivot column. The first entry in the first row, 1, is the first pivot entry and we will proceed with the rest.

LINEAR SYSTEM OF EQUATIONS

$$\left(\begin{array}{ccc|c} 1 & 4 & 3 & 11 \\ 2 & 10 & 7 & 27 \\ 1 & 1 & 2 & 5 \end{array}\right) \xrightarrow{\leftarrow R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & 4 & 3 & 11 \\ 0 & 2 & 1 & 5 \\ 1 & 1 & 2 & 5 \end{array}\right) \xrightarrow{\leftarrow R_3 - R_1} \left(\begin{array}{ccc|c} 1 & 4 & 3 & 11 \\ 0 & 2 & 1 & 5 \\ 0 & -3 & -1 & -6 \end{array}\right)$$

- Changing the last row to zeros except the last entry which changes to 1.

$$\left(\begin{array}{ccc|c} 1 & 4 & 3 & 11 \\ 0 & 2 & 1 & 5 \\ 0 & -3 & -1 & -6 \end{array}\right) \xrightarrow{\leftarrow 2R_3} \left(\begin{array}{ccc|c} 1 & 4 & 3 & 11 \\ 0 & 2 & 1 & 5 \\ 0 & -6 & -2 & -12 \end{array}\right) \xrightarrow{\leftarrow R_3 + 3R_2} \left(\begin{array}{ccc|c} 1 & 4 & 3 & 11 \\ 0 & 2 & 1 & 5 \\ 0 & 0 & 1 & 3 \end{array}\right)$$

- Changing the remaining pivot columns to zeros in row 2.

$$\left(\begin{array}{ccc|c} 1 & 4 & 3 & 11 \\ 0 & 2 & 1 & 5 \\ 0 & 0 & 1 & 3 \end{array}\right) \xrightarrow{\leftarrow R_2 - R_3} \left(\begin{array}{ccc|c} 1 & 4 & 3 & 11 \\ 0 & 2 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array}\right) \xrightarrow{\leftarrow \frac{1}{2} R_2} \left(\begin{array}{ccc|c} 1 & 4 & 3 & 11 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 3 \end{array}\right)$$

LINEAR SYSTEM OF EQUATIONS

- Let's finish with row 1, make 0's on all pivot columns

$$\left(\begin{array}{ccc|c} 1 & 4 & 3 & 11 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 3 \end{array}\right) \xrightarrow{\leftarrow R_1 - 3R_3} \left(\begin{array}{ccc|c} 1 & 4 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 3 \end{array}\right) \xrightarrow{\leftarrow R_1 - 4R_2} \left(\begin{array}{ccc|c} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 3 \end{array}\right)$$

- Since all the rows have been entirely reduced, each pivot holds a value of one variable.

- Hence, $\bar{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 3 \end{bmatrix}$; in other words $x_1 = -2$, $x_2 = 1$ and $x_3 = 3$.

LINEAR SYSTEM OF EQUATIONS

Example 5: Solve the following equations: $3x_1 - x_2 + 5x_3 = 8$

$$x_2 - 10x_3 = 1$$

$$6x_1 - x_2 = 17$$

$$\left(\begin{array}{ccc|c} 3 & -1 & 5 & 8 \\ 0 & 1 & -10 & 1 \\ 6 & -1 & 0 & 17 \end{array}\right) \xrightarrow{\;\;\;} \left(\begin{array}{ccc|c} 3 & -1 & 5 & 8 \\ 0 & 1 & -10 & 1 \\ 0 & 1 & -10 & 1 \end{array}\right) \leftarrow R_3 - 2R_1 \xrightarrow{\;\;\;} \left(\begin{array}{ccc|c} 3 & -1 & 5 & 8 \\ 0 & 1 & -10 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right) \leftarrow R_3 - R_2$$

- This is the echelon form not the reduced echelon form. Converting this to reduced echelon form will consists of multiplying row 1 by $\frac{1}{3}$ hence, we will have fractions. In this form, we can deduce equations that satisfy the given equation solutions.

LINEAR SYSTEM OF EQUATIONS

- In the echelon form,

Pivots

$$\begin{array}{cccc} x_1 & x_2 & x_3 & b \\ \end{array}$$
$$\left(\begin{array}{ccc|c} 3 & -1 & 5 & 8 \\ 0 & 1 & -10 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

- Variables that are represented by pivot columns are called pivot variables and the variables that aren't represented by pivot columns are called free variables.

- In this case, x_1 and x_2 are pivot variables and x_3 is a free variable.

LINEAR SYSTEM OF EQUATIONS

- Thus, our solution to the equations include

$$3x_1 - x_2 + 5x_3 = 8$$
$$x_2 - 10x_3 = 1$$

- We may notice that x_3 is not constrained by any other equation, that is, we can let x_3 to be equal to any number. If we let x_3 to be equal to t, t will be known as a parameter. Thus, $x_2 - 10t = 1; x_2 = 1 + 10t$ and $3x_1 - (1 + 10t) + 5t = 8; x_1 = 9 + \frac{5}{3}t$ (infinite number of solutions)

- Therefore, $x_1 = 9 + \frac{5}{3}t$, $x_2 = 1 + 10t$, $z = t$. If we let the value of t to be 4, we get the following values: $x_1 = \frac{29}{3}, x_2 = 41$ and $x_3 = 4$.

LINEAR SYSTEM OF EQUATIONS

- Consider the following equations;

$$x + 2y - 2z + 2w = 7$$
$$x + 2y - z + 3w = 5$$
$$x + 2y - 3z + w = 1$$

- The augmented matrix and its reduced echelon form are

$$\left(\begin{array}{cccc|c} 1 & 2 & -2 & 2 & 3 \\ 1 & 2 & -1 & 3 & 5 \\ 1 & 2 & -3 & 1 & 1 \end{array}\right) \xrightarrow{\text{RREF}} \left(\begin{array}{cccc|c} 1 & 2 & 0 & 2 & 7 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

- Thus, $\bar{x} = \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 7 - 2y - 2w \\ y \\ 2 - w \\ w \end{bmatrix} = \begin{bmatrix} 7 - 2t - 2s \\ t \\ 2 - s \\ s \end{bmatrix}$ for $y = t, w = s$ and $\forall\{y, t\} \in \mathbb{R}$.

LINEAR SYSTEM OF EQUATIONS

- The following are the properties of a linear system of equations.
  1. No solution: If the echelon form has a row of the form $(0 \quad 0 \quad 0|b)$, where $b \neq 0$, then the system is inconsistent and has no solution.
  2. One solution: If every column of the coefficient matrix of the echelon form is a pivot column, the system has exactly one solution.
  3. Infinitely many solutions: For a consistent system of equations: If not all columns of the coefficient matrix of the echelon form are pivot columns, then the system has infinitely many solutions.

Example 6: For homogenous, solve the following system of equations.

$$2x_1 + x_2 + x_3 + 4x_4 = 0$$
$$x_1 + 2x_2 - x_3 + 5x_4 = 0$$

$$\left(\begin{array}{cccc|c} 2 & 1 & 1 & 4 & 0 \\ 1 & 2 & -1 & 5 & 0 \end{array}\right) \xrightarrow{\text{RREF}} \left(\begin{array}{cccc|c} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & 2 & 0 \end{array}\right)$$

- According to the reduced echelon form,

$$x_1 = -x_3 - x_4 \text{ and } x_2 = x_3 - 2x_4$$

That is, $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -x_3 - x_4 \\ x_3 - 2x_4 \\ x_3 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} -1 \\ 1 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -1 \\ -2 \\ 0 \\ 1 \end{bmatrix}$

LINEAR SYSTEM OF EQUATIONS

- Notice that we have constructed a column from the coefficients of x_3 in each equation, and another column from the coefficients of x_4.
- Consider what happens when we let the parameters to be $x_3 = 1$ and $x_4 = 0$;

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = 1\begin{bmatrix} -1 \\ 1 \\ 1 \\ 0 \end{bmatrix} + 0\begin{bmatrix} -1 \\ -2 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

- And if we let $x_3 = 0$ and $x_4 = 1$;

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = 0\begin{bmatrix} -1 \\ 1 \\ 1 \\ 0 \end{bmatrix} + 1\begin{bmatrix} -1 \\ -2 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \\ 0 \\ 1 \end{bmatrix}$$

These two vector are called basic solutions. We say that the general solution of the homogeneous system is a linear combination of its basic solutions

LINEAR SYSTEM OF EQUATIONS

Question: Solve the following systems of linear equations.

1. $2x_1 + 3x_2 + 4x_3 = 0$

   $x_1 - 3x_2 + x_3 = 0$

   $4x_1 - x_2 + 6x_3 = 0$

2. $x + \mathrm{y} - \mathrm{z} + 2\mathrm{w} = 0$

   $x + 3y + z + 6w = 0$

   $x + 2y + 4w = 0$

3. $-x_2 + x_3 = 3$

   $x_1 - x_2 - x_3 = 0$

   $-x_1 - x_2 = -3$

4. $x_2 - 4x_2 = 8$

   $2x_1 - 3x_2 + 2x_3 = 1$

   $4x_1 - 8x_2 + 12x_3 = 1$

ROW SPACE, COLUMN SPACE AND NULL SPACE

ROW SPACE, COLUMN SPACE AND NULL SPACE

- In this section we will study some important vector spaces that are associated with matrices. Our work here will provide us with a deeper understanding of the relationships between the solutions of a linear system and properties of its coefficient matrix.
  - Row space: This refers to the space spanned by the row vectors of a matrix. In other words, it's the set of all possible linear combinations of the rows of a matrix.
  - Column space: It's the space spanned by the column vectors of a matrix. Like the row space, it's the set of all possible linear combinations of the columns.
  - Null space: This is the set of all vectors that satisfy the homogenous equations. In essence, it represents the solutions to the equation $A\bar{x} = 0$, where A is the matrix and $\bar{x}$ is a vector of appropriate dimension.

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- For an $m \times n$ matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

- The vectors $r_1 = [a_{11}\ a_{12}\ \cdots\ a_{1n}]$, $r_2 = [a_{21}\ a_{22}\ \cdots\ a_{2n}]$ and so on up to $r_m = [a_{m1}\ a_{m2}\ \cdots\ a_{mn}]$ in R^n that are formed from the rows of A are called the *row vectors* of A.
- This implies that the vectors R^m formed from the columns of A are called the *column vectors* of A.

ROW SPACE, COLUMN SPACE AND NULL SPACE

- From the given matrix A, below are the column vectors.

$$c_1 = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}, c_2 = \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} \text{ and so on up to } c_3 = \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}$$

Definition: Column space, row space, null space

Let A be an $m \times n$ matrix. The column space of A, written col(A), is the span of the columns. The row space of A, written row(A), is the span of the rows. The null space of A, written null(A), is the set

$$\text{null}(A) = \{\bar{x} \mid A\bar{x} = 0\}.$$

ROW SPACE, COLUMN SPACE AND NULL SPACE

- Now, we may note that the product $A\bar{x}$, in the equation $A\bar{x} = 0$, can be written as a linear combination of the row and column matrix consisting of coefficients from $\bar{x}$. That is

$$A\bar{x} = x_1 c_1 + x_2 c_2 + \cdots + x_n c_n$$

- Thus, a linear system, $A\bar{x} = \bar{b}$, of m equations in n unknowns can be written as $x_1 \boldsymbol{c_1} + x_2 \boldsymbol{c_2} + \cdots + x_n \boldsymbol{c_n} = \bar{b}$ from which we conclude that $A\bar{x} = \bar{b}$ is consistent if and only if $\bar{b}$ is expressible as a linear combination of the column vectors of A.

Example 7: A Vector $\bar{b}$ in the Column Space of A

Let $A\bar{x} = \bar{b}$ be the linear system

$$\begin{bmatrix} -1 & 3 & 2 \\ 1 & 2 & -3 \\ 2 & 1 & -2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -9 \\ -3 \end{bmatrix}$$

Show that $\bar{b}$ is in the column space of A by expressing $\bar{b}$ as a linear combination of column vectors of A.

- Solving this system using Gaussian Elimination method gives $x_1 = 2$, $x_2 = -1$ and $x_3 = 3$ *(please verify this)*

- Since $c_1 = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$, $c_2 = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$, $c_3 = \begin{bmatrix} 2 \\ -3 \\ -2 \end{bmatrix}$ and $x_1\boldsymbol{c_1} + x_2\boldsymbol{c_2} + x_3\boldsymbol{c_3} = \bar{\mathrm{b}}$,

Thus,

$$x_1 \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} 2 \\ -3 \\ -2 \end{bmatrix} = 2 \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} 2 \\ -3 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ -9 \\ -3 \end{bmatrix}$$

ROW SPACE, COLUMN SPACE AND NULL SPACE

- We know that performing elementary row operations on the augmented matrix $[A\,|\bar{b}]$ does not change the solution set of that system. This is also true if the system is homogeneous, where the augmented matrix is $[A \mid 0]$.

Example 8: Finding a Basis for the Null Space of a Matrix

Find the basis for null space of the matrix

$$\left(\begin{array}{cccccc|c} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 2 & 6 & -5 & -2 & 4 & -3 & 0 \\ 0 & 0 & 5 & 10 & 0 & 15 & 0 \\ 2 & 6 & 0 & 8 & 4 & 18 & 0 \end{array}\right)$$

- After the row operations, we get the following matrix in RREF.

$$\left(\begin{array}{cccccc|c} 1 & 3 & 0 & 4 & 2 & 0 & 0 \\ 0 & 0 & 1 & -2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

- This results into the following equations;

$$x_1 = -3x_2 - 4x_4 - 2x_5$$

$$x_3 = -2x_4$$

$$x_6 = 0$$

- This means that x_1, x_3 and x_6 are pivot variables while x_2, x_4 and x_5 are free variables.

- Therefore, $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} -3x_2 - 4x_4 - 2x_5 \\ x_2 \text{ is free} \\ -2x_4 \\ x_4 \text{ is free} \\ x_5 \text{ is free} \\ 0 \end{bmatrix}$ and if we factorise;

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = x_2 \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -4 \\ 0 \\ -2 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -2 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

- Therefore;

$$u = \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, v = \begin{bmatrix} -4 \\ 0 \\ -2 \\ 0 \\ 0 \\ 0 \end{bmatrix} \text{ and } w = \begin{bmatrix} -2 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

Basis

- Observe that the basis vectors u,v, and w in this example are the vectors that result by successively setting one of the parameters (free variables) in the general solution equal to 1 and the others equal to 0.

Example 9: Finding the Null Space of a Matrix

Find a spanning set for the null space of the matrix

$$\begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

- The first step is to find the general solution of $A\bar{x} = 0$ in terms of free variables. The reduced echelon form of the augmented matrix is

$$\left(\begin{array}{ccccc|c} 1 & -2 & 0 & -1 & 3 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

- At this point we need to write the pivot variables in terms of the free variables.

$$x_1 = 2x_2 + x_4 - 3x_5$$
$$x_3 = -2x_4 + 2x_5$$

- x_1 and x_3 are the only pivot variables. The rest are free.

ROW SPACE, COLUMN SPACE AND NULL SPACE

- We now decompose the vector giving the general solution into a linear combination of vectors.

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2x_2 + x_4 - 3x_5 \\ x_2 \text{ is free} \\ -2x_4 + 2x_5 \\ x_4 \text{ is free} \\ x_5 \text{ is free} \end{bmatrix} = x_2 \overset{\overset{u}{\downarrow}}{\begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}} + x_4 \overset{\overset{v}{\downarrow}}{\begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix}} + x_5 \overset{\overset{w}{\downarrow}}{\begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}}$$

- Thus, $\text{Null(A)} = \{u, v, w\} = \left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right\}$

Example 10: Basis of column space and row space

Find a basis for the column space, row space of the matrix

$$\begin{bmatrix} 1 & 2 & 1 & 3 & 2 \\ 1 & 3 & 6 & 0 & 2 \\ 3 & 7 & 8 & 6 & 6 \end{bmatrix}$$

ROW SPACE, COLUMN SPACE AND NULL SPACE

- The column space is the span of the columns of the matrix A, i.e.;

$$\text{Col(A)=Span}\left\{\begin{bmatrix}1\\1\\3\end{bmatrix},\begin{bmatrix}2\\3\\7\end{bmatrix},\begin{bmatrix}1\\6\\8\end{bmatrix},\begin{bmatrix}3\\0\\6\end{bmatrix},\begin{bmatrix}2\\2\\6\end{bmatrix}\right\}$$

- The row-reduced echelon for of A is $\begin{bmatrix}1 & 0 & -9 & 9 & 2\\0 & 1 & 5 & -3 & 0\\0 & 0 & 0 & 0 & 0\end{bmatrix}$ and the pivot column in the RREF of A represent the basis pf Col(A)

$$\text{Basis of Col(A)} = \left\{\begin{bmatrix}1\\1\\3\end{bmatrix},\begin{bmatrix}2\\3\\7\end{bmatrix}\right\}$$

ROW SPACE, COLUMN SPACE AND NULL SPACE

- Row space of A is the span of the rows, i.e.;

$$\text{Row(A)} = \text{span}\{[1 \quad 2 \quad 1 \quad 3 \quad 2], [1 \quad 3 \quad 6 \quad 0 \quad 2], [3 \quad 7 \quad 8 \quad 6 \quad 6]\}$$

- Non-zero rows of the row-reduced echelon form of the given matrix gives the basis for the row space of A.
- Therefore;

$$\text{Basis of Row(A)} = \{[1 \quad 0 \quad -9 \quad 9 \quad 2], [0 \quad 1 \quad 5 \quad -3 \quad 0]\}$$

Question: Find null(A) for the following matrices.

1. $A = \begin{bmatrix} 2 & 3 \\ 4 & 6 \end{bmatrix}$

2. $A = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 3 \\ 3 & 2 & 1 \end{bmatrix}$

3. $A = \begin{bmatrix} 2 & -1 & 3 & 5 \\ 2 & 0 & 1 & 2 \\ 6 & 4 & -5 & -6 \\ 0 & 2 & -4 & -6 \end{bmatrix}$

4. $A = \begin{bmatrix} 1 & 3 & 5 & 0 \\ 0 & 1 & 4 & 2 \end{bmatrix}$

5. $A = \begin{bmatrix} 1 & 5 & -4 & -3 & 1 \\ 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

Thank You For Reading!!

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